VICTORY 2027

PHOTOSYNTHESIS
(PHOTOSYNTHESIS PROCESS)

LECTURE-01 | BIOLOGY

By– Upasana Ma'am



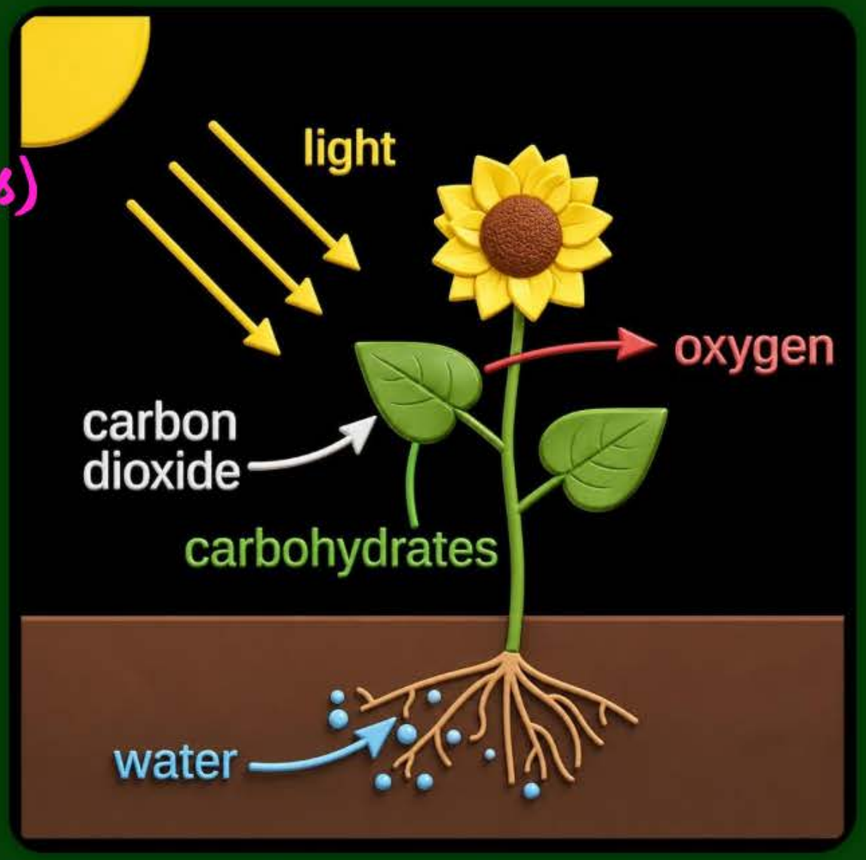
TOPICS TO BE COVERED

1. What is photosynthesis
2. Importance of Photosynthesis
3. Chlorophyll- The vital plant pigment
4. Chloroplast
5. Regulation of stomatal opening for letting in carbon dioxide
6. K^+ ion concentration theory
7. [redacted]


HELLO
NINJAS
忍者参上
影の中で戦う
忍者
忍術最強
忍者
忍者参上

What is Photosynthesis?

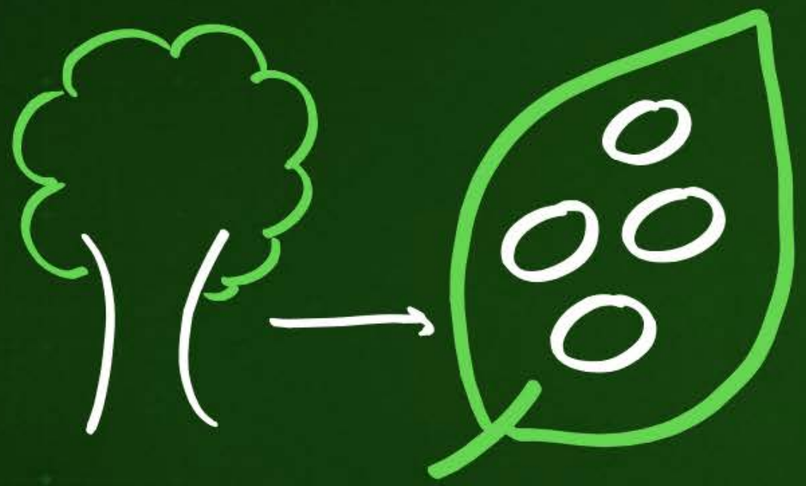
- Photosynthesis is the process by which green plants prepare their own food. (Autotrophs)
- It takes place in plant cells that contain chlorophyll.
- During this process, plants use carbon dioxide, water, and light energy to make food in the form of glucose, which is later stored as starch. (Stored form of Glucose)
- Oxygen is released as a by-product during photosynthesis

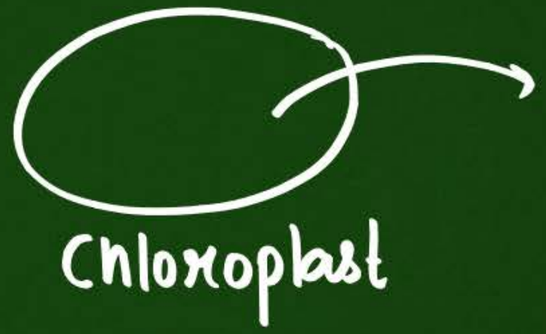


$$\underline{6}CO_2 + 12H_2O \xrightarrow[\text{chlorophyll}]{\text{sunlight}} \underline{C_6H_{12}O_6}_{\text{Glucose}} + 6O_2\uparrow + 6H_2O$$

Reactants

By-product





Chloroplast

Chlorophyll

green coloured pigment

'Photosynthesis'

light to make

X



DUGGU

↓

Heterotrophic

Different

↳ Done by Green plants

(Autotrophs)

Self trophic → Nutrition



Importance of Photosynthesis

1. Food for All :

❖ Photosynthesis is ultimately the source of energy and food for all living beings- directly for plants and indirectly for animals and humans who eat the plants or the plant-eating animals.

2. Oxygen to Breathe in:

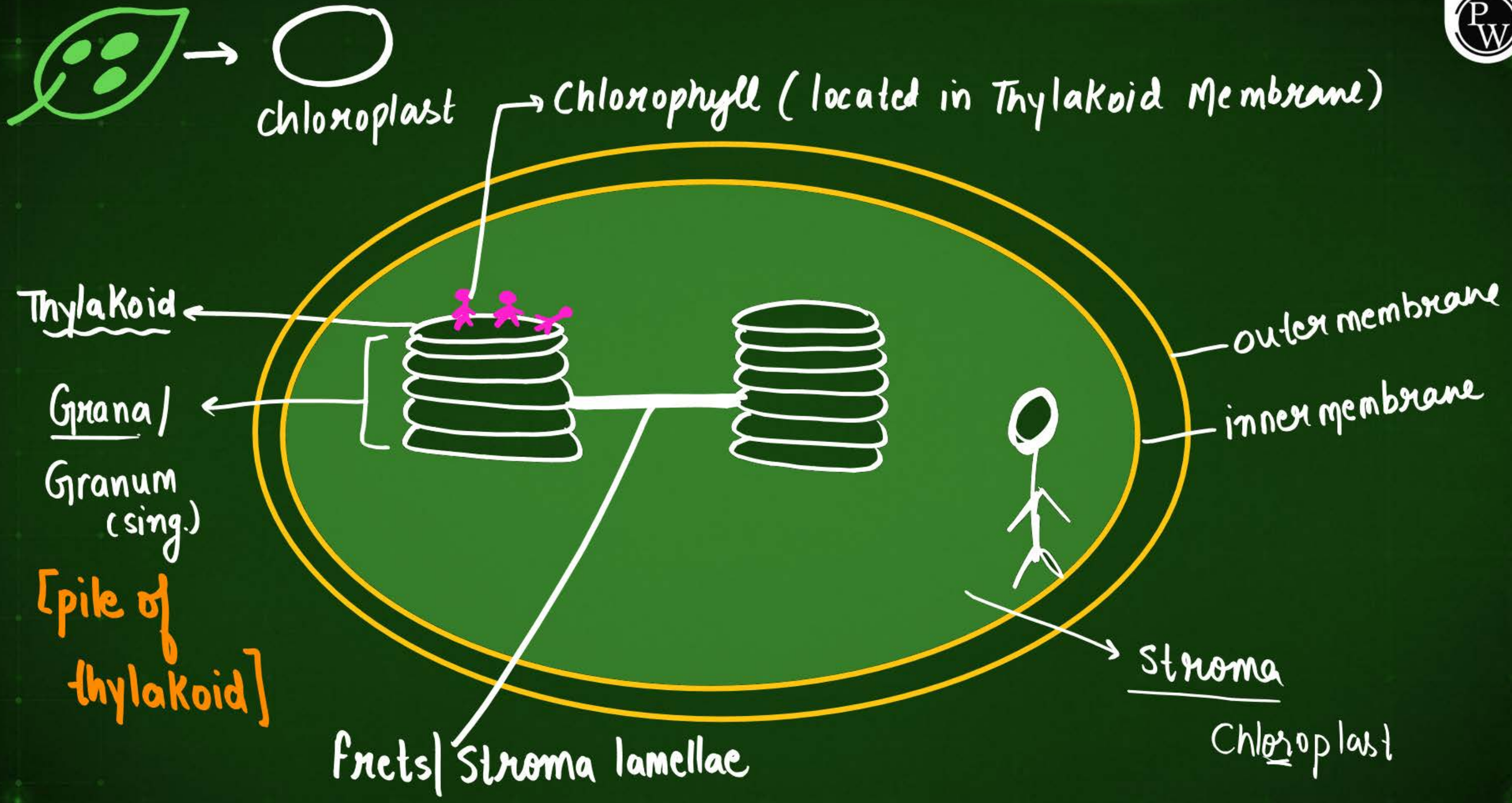
- Photosynthesis is the only biological process which releases oxygen into the atmosphere. Oxygen supports all life on earth.

Chlorophyll- The Vital Plant Pigment

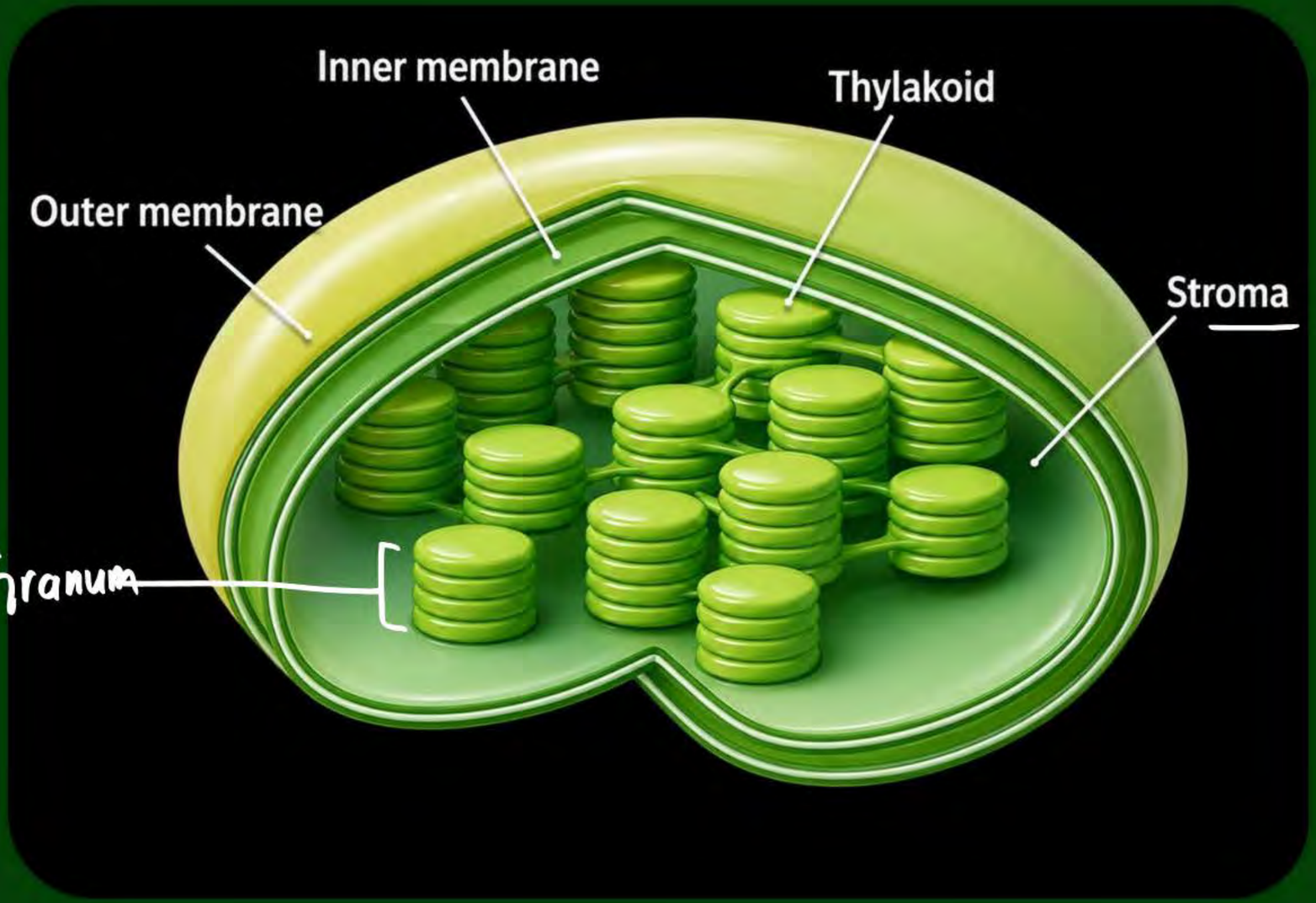
- **Chlorophyll** (chloro : green, phyll : leaf) is the green pigment found in plants.
- These are small, oval-shaped structures surrounded by a double membrane.
- Inside the chloroplast are flattened sacs called thylakoids.
- Thylakoids are arranged in stacks called grana
- These grana are present in a colourless fluid called the stroma.
- Chlorophyll, the green pigment, is present in the walls of the thylakoids.
- Chlorophyll is a complex substance made up of carbon, hydrogen, oxygen, nitrogen, and magnesium.

membrane

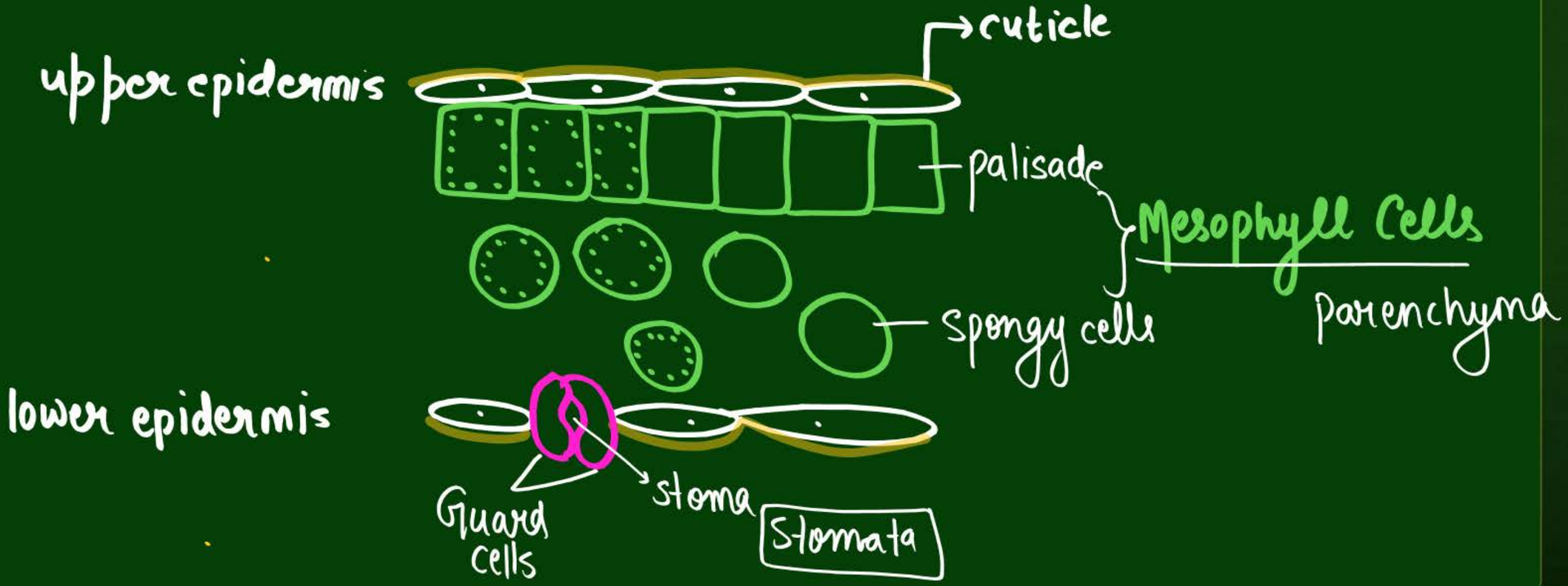
Magi CHON hai?
↓
Magnesium


Chloroplast
Chlorophyll (located in Thylakoid Membrane)
Thylakoid
Grana / Granum (sing.)
[pile of thylakoid]
outer membrane
inner membrane
Stroma
Chloroplast
Frets / Stroma lamellae

Chloroplast



- Chloroplasts are mainly contained in the mesophyll cells located between the upper epidermis and the lower epidermis (i.e., in palisade cells and spongy cells) of leaves. These are also found in the guard cells of stomata and in the outer layers of young green stems.



Some Facts

VIBGYOR

- Chlorophylls absorb light at both ends of the visible spectrum - i.e., BLUE and RED light and reflect away the green light. That is why, chlorophyll appears green. The absorbed blue and red lights are most effective for photosynthesis.

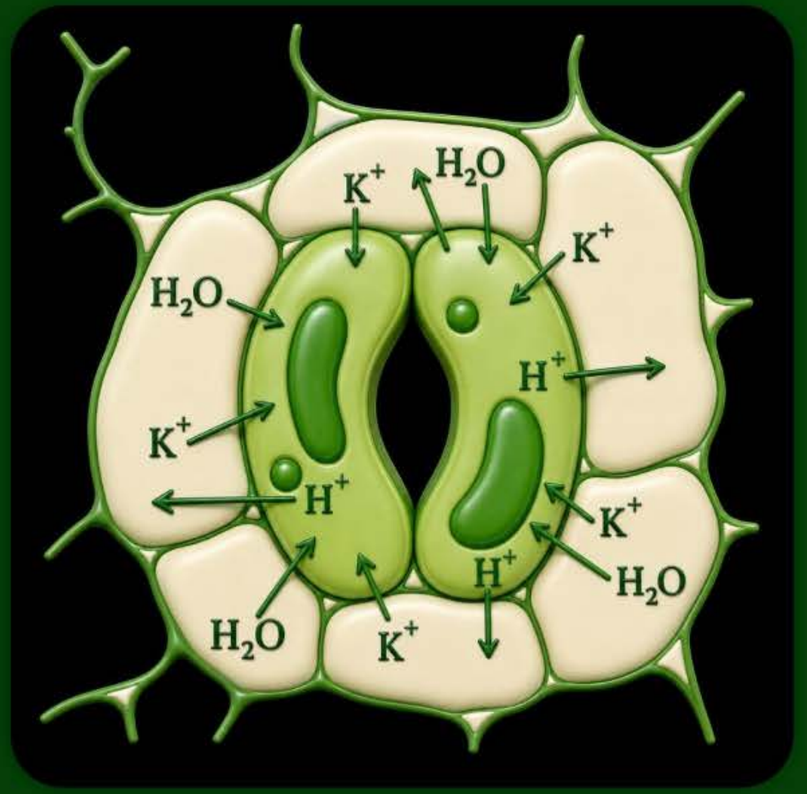
Chlorophyll 'a' and 'b' are most abundant.

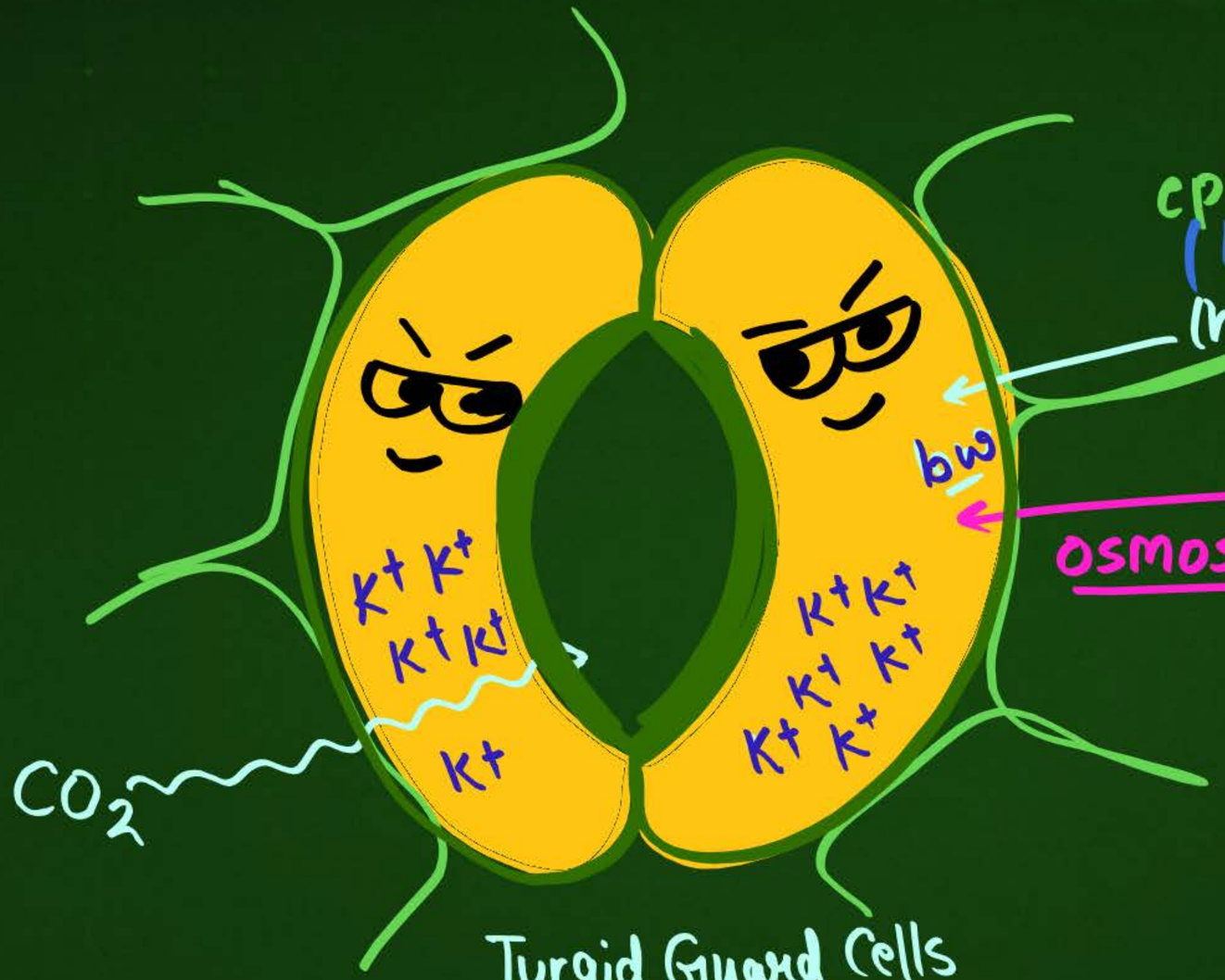
K^+ Ion Concentration Theory

Opening of Stomata

Active Transport

- Guard cells uses ATP actively to pump the potassium ions of epidermal cells
- Due to the high K^+ concentration, water enters the guard cells by osmosis.
- The guard cells become turgid (swollen).The swollen guard cells bend outward, and the stomata open.





Turgid Guard Cells during day

Guard Cells open during day.

but close at Night



Sugar Concentration Theory

- During daytime, the guard cells begin photosynthesis and the sugar (glucose) produced during the process increases the osmotic pressure which draws in water from the adjoining cells due to endosmosis. Hence, the guard cells become turgid and bulge outwards.

QUESTION

Answer the following in "Yes" or "No".

(i) All parts of a green plant carry out photosynthesis.

1. Yes
2. No

QUESTION

(ii) All green parts of a plant carry out photosynthesis.

↓

Chloxophyll

1. Yes
2. No

2023

Arrange in the logical sequence.

Snake, Rabbit, Cabbage, Hawk

S 3 R 2 C 1 H 4

A) C, H, S, R
B) C, R, S, H
C) C, S, R, H

QUESTION

(iii) Photosynthesis is the only biological process that releases oxygen into the air.

1. Yes ✓
2. No

QUESTION

(iv) Out of nine types of chlorophylls, chlorophyll a and b are the most abundant.

1. Yes
2. No

QUESTION

(v) Too much light destroys chlorophyll.

1. Yes ✓
2. No

QUESTION

H.W

(vi) No transpiration occurs during photosynthesis.

↓

Guard cells open

↓

CO_2 inside.

1. Yes — True
2. No — False

Thank You